Entity Relationship Modelling

Components of an ERD:

- Entity – tables in the database
- Attributes – column in the table
- Relationship: the association between entities: one to one, one to many, many to many
- Constraints – restriction / rules enforced on the data in the database (example: PK, FK)

1 to 1 relationship

- Example: Manager manages a department, department managed by one manager
- Can place the FK in either one / anyone of the table (but not both)
- However, it's better to place the FK in the table which is mandatory in the relationship. For example, if department must have a manager but manger not necessary must manage a department, then place the FK in department table.

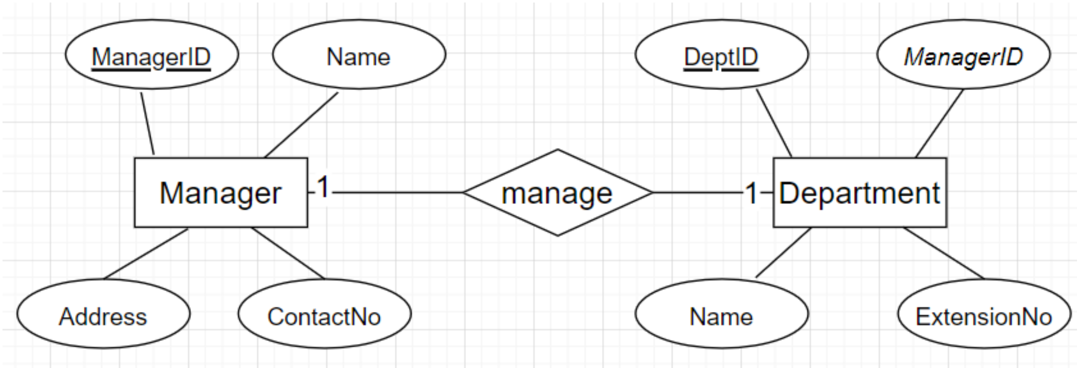
Manager

ManagerID (PK)	Name	Address	ContctNo
M01	Ali	Shah Alam	010123456
M02	Ahmed	Kuala Lumpur	010126456
M03	Jason	Puchong	010823456
M04	Helen	Sri Petaling	010123496

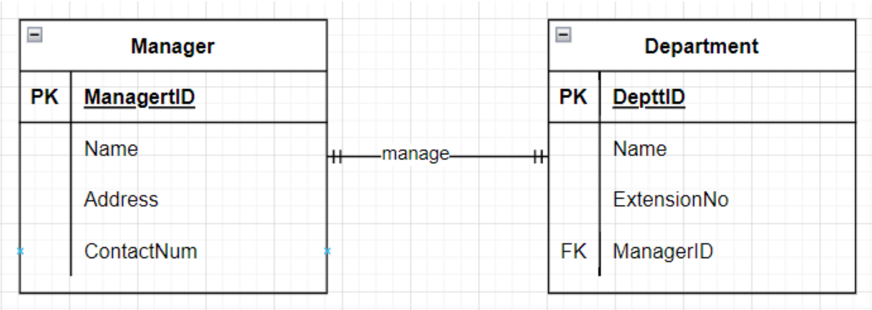
Department

DeptID (PK)	Name	ExtensionNo	ManagerID (FK)
D01	Sales	101	M02
D02	Marketing	102	M01
D03	IT	103	M03

ERD Chen's Notation



ERD Crow's Foot Notation



The association between PK/FK is always 'one to one' match

Each DeptID is associated to 1 ManagerID & Each ManagerID is associated to 1 DeptID

ManagerID (one)	(one) **DeptID**
M01	D02
M02	D01
M03	D03
M04	

<u>1 to Many relationship</u>

- Example: Lecturer supervises many students, student supervised by one lecturer
- Must place the FK in the 'many' table which is student table, because lecturer supervises 'many' students

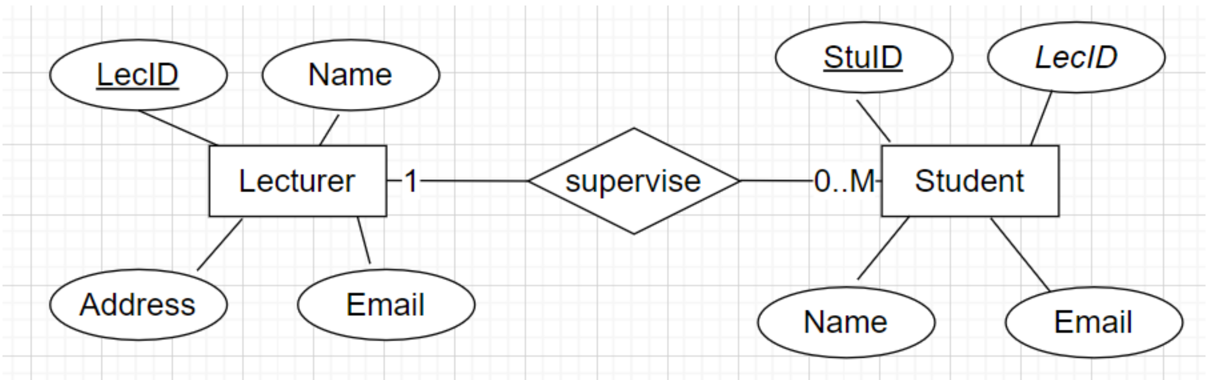
Lecturer

LecID (PK)	Name	Address	Email
L01	Ali	Shah Alam	abc@gmail.com
L02	Kelly	Kuala Lumpur	dfg@gmail.com
L03	John	Puchong	hkj@gmail.com
L04	William	Sri Petaling	rty@gmail.com

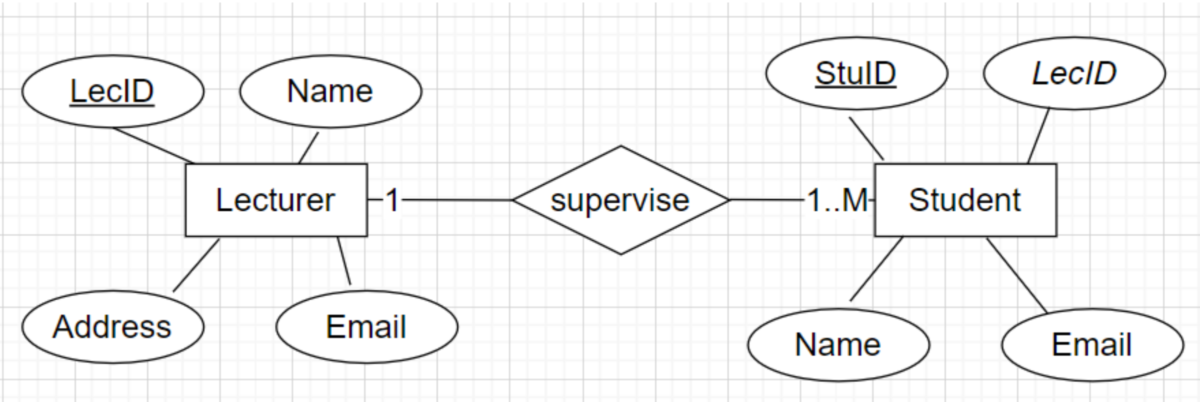
Student

StuID (PK)	Name	Email	LecID (FK)
S01	Nelson	abc@yahoo.com	L02
S02	Melvin	dfg@yahoo.com	L02
S03	Oscar	hkj@yahoo.com	L01
S04	Peter	rty@yahoo.com	L03
S05	Rachel	jkl@yahoo.com	L01

ERD Chen's Notation

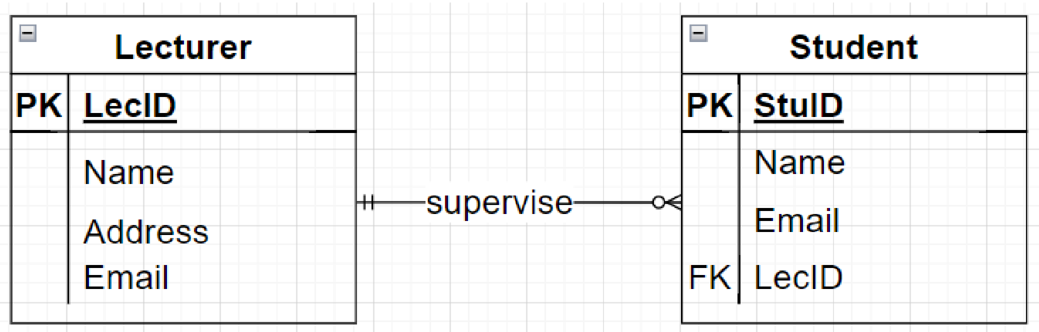


'zero/one/many' on student's side, meaning student is optional to Lecturer. Some lecturers may supervise zero student, some lecturers may supervise one student, some lecturers may supervise many students.

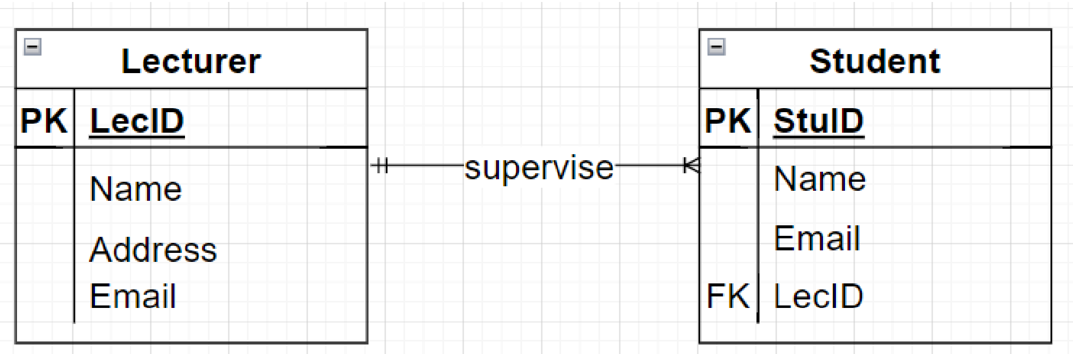


'one/many' on student's side, meaning student is mandatory to Lecturer. Each lecturer supervises at least one or many students.

ERD Crow's Foot Notation



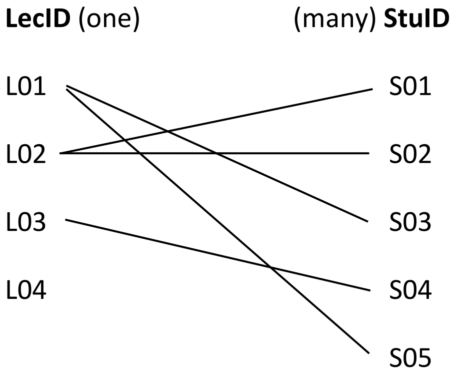
'zero/one/many' on student's side, meaning student is optional to Lecturer. Some lecturers may supervise zero student, some lecturers may supervise one student, some lecturers may supervise many students.



'one/many' on student's side, meaning student is mandatory to Lecturer. Each lecturer supervises at least one or many students.

The association between PK & FK

Each StuID is associated to 1 LecID, but each LecID can be associated to more than 1 StuID



*Some lecturers might supervise zero or one student, but as long as there's a lecturer who supervises more than one student, then the relationship is consider as many (more than 1 is consider as many)

<u>Many to many relationship</u>

- Example: Supplier supplies many products, product supplied by many suppliers
- In this case, you need a bridge entity / associative entity (because the relationship is many to many)
- An extra table is needed in between supplier and product
- **Place the FK in the bridge entity**

Supplier

SupplierID (PK)	Name	Address	ContactNo
S01	ABC Company	KL	010123456
S02	XYZ Company	PJ	010126456
S03	HJK Company	PJ	010823456
S04	PQR Company	JB	010123496

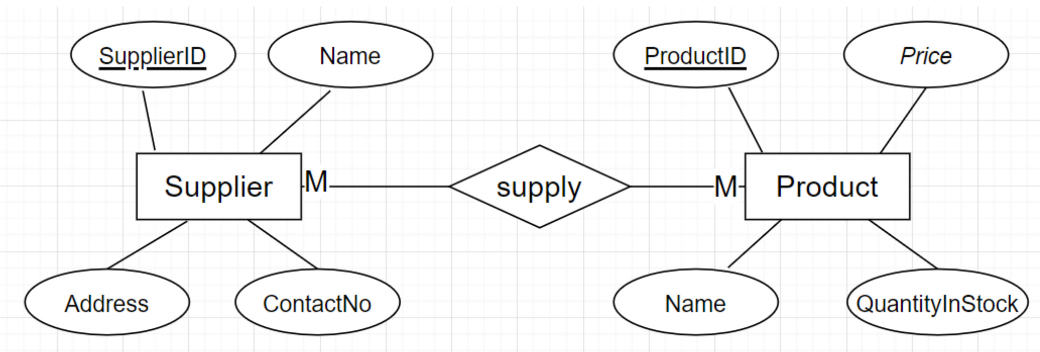
Product

ProductID (PK)	Name	Price	QuantityInStock
P01	Keyboard	RM80	100
P02	Mouse	RM30	120
P03	Monitor	RM200	80

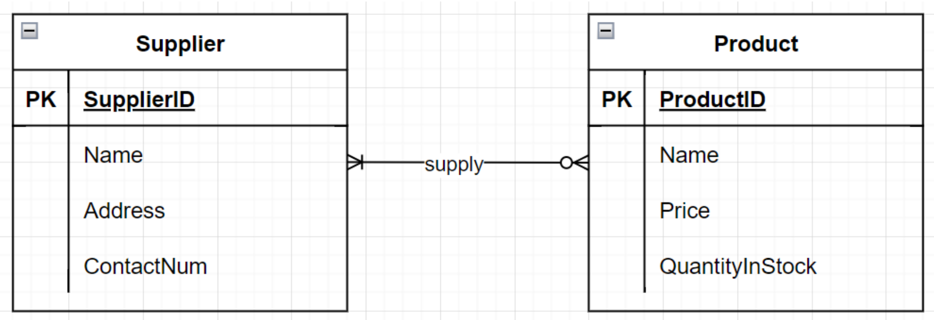
Supplies (bridge entity / associative entity)

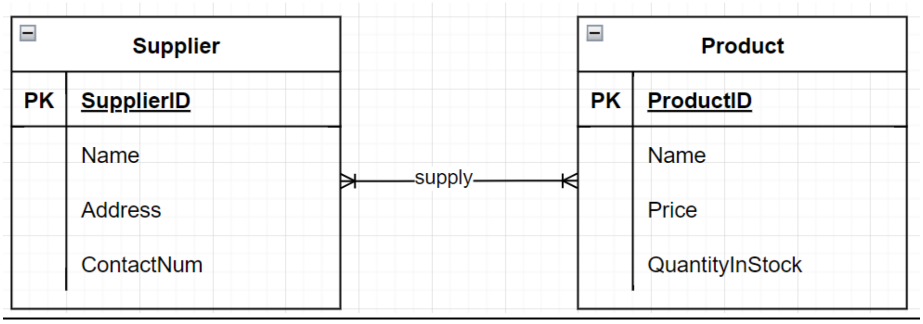
SuppliesID(PK)	SupplierID (FK)	ProductID (FK)	SuppliedDate	QuantitySupplied
SU01	S01	P01	1 Jan	200
SU02	S01	P02	2 Jan	250
SU03	S01	P03	2 Feb	190
SU04	S02	P03	3 Mar	280
SU05	S03	P02	1 April	300

ERD Chen's Notation (without showing the bridge entity)



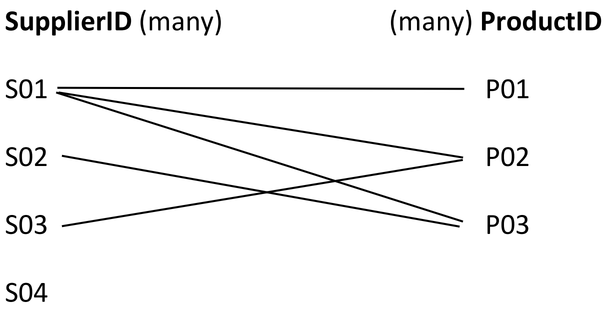
ERD Crow's Foot Notation (without showing the bridge entity)





The association between PK/FK of Supplier & Product table is 'many to many'

Each SupplierID is associated to more than one ProductID & Each ProductID is associated to more than 1 SupplierID



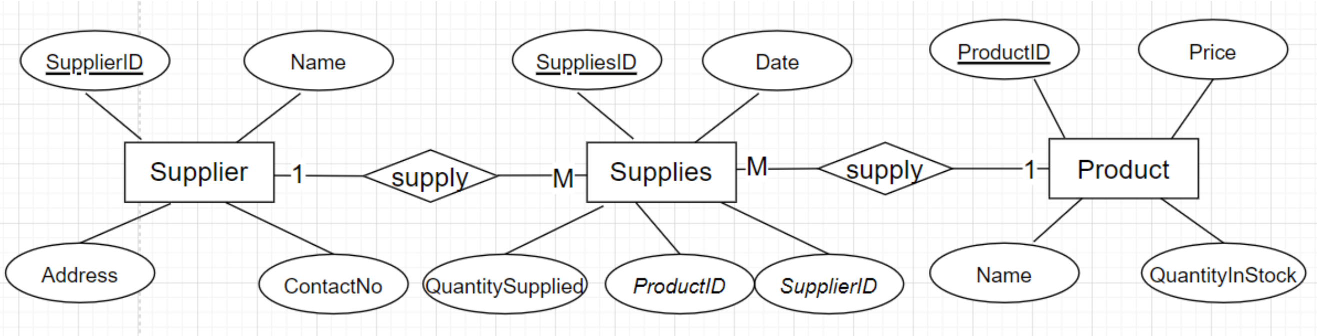
*Some suppliers might supply zero or one product, but as long as there's a supplier who supply more than one product, then the relationship is consider as many (more than 1 is consider as many)

*Some products might be supplied by one supplier, but as long as there's a product which is supplied by more than one supplier, then the relationship is consider as many (more than 1 is consider as many)

*However, u cannot have a product which is not supplied by any supplier because this is not logic (unless it's being specified in the business rules)

When drawing the ERD, it is **better** to show the bridge entity because it will be more detailed, in such case, the 'many to many' relationship is 'resolved such as below

ERD Chen's Notation



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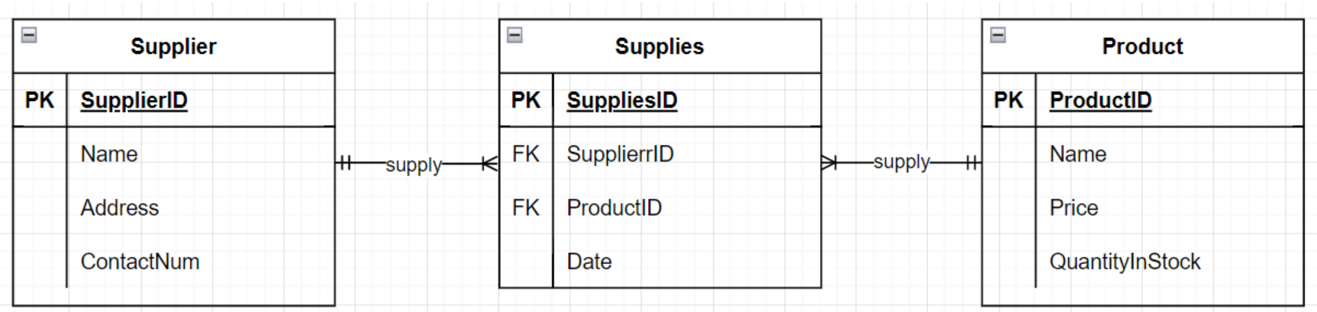


Each SupplierID is associated to 1 or more SuppliesID, but each SuppliesID is associated to 1 SupplierID

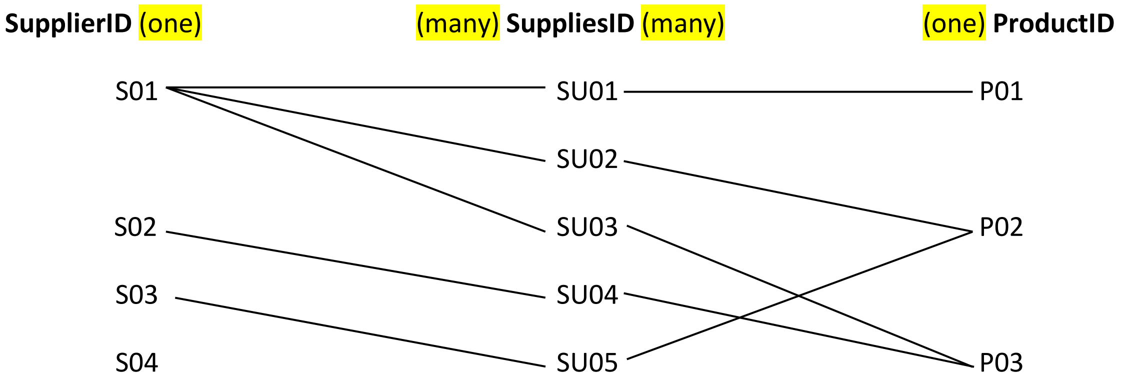
Each ProductID is associated to 1 or more SuppliesID, but each SuppliesID is associated to 1 ProductID

Resolving 'Many to Many' relationship into TWO '1 to Many' relationship

- **This happens when we show the bridge entity in the ERD**
- **Now, the relationship between Supplier and Supplies (bridge entity) is 'one to many'**
- **And the relationship between Product and Supplies (bridge entity) is 'one to many'**
- **But the relationship between Supplier and Product is STILL 'many to many'**
- **The bridge entity's side will be 'many' while the other side will be 'one'**

Supplies (bridge entity / associative entity)

SuppliesID(PK)	SupplierID (FK)	ProductID (FK)	Date	QuantitySupplied
SU01	S01	P01	1 Jan	200
SU02	S01	P02	2 Jan	250
SU03	S01	P03	2 Feb	190
SU04	S02	P03	3 Mar	280
SU05	S03	P02	1 April	300



*Bridge entity's side is 'many' because each SupplierID and ProductID may appear many times in the bridge entity. But each SuppliesID is only associated to one SupplierID and ProductID.